Cake Cutting Algorithms
Fair Division Approach
A Berkeley Splash Class

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April 13, 2025

Introduction

- Have you ever had to share a cake with friends?
- How do you make sure everyone gets a "fair" piece?
- What does "fair" even mean?

Today's Objectives

By the end of this class, you will:

- Understand basic cake cutting algorithms
- Know how to apply these to real situations
- Learn about cake numbers and their mathematical significance
- See connections to other areas of mathematics and computer science

What is Fair Division?

- The mathematical study of how to divide resources fairly
- Key concepts:
  - **Proportionality**: Everyone gets at least 1/n of the total value
  - **Envy-freeness**: No one prefers someone else's piece
  - **Efficiency**: No better allocation exists

Simplest Case: Two People — "I Cut, You Choose"

- One person cuts the cake into two pieces they consider equal
- The other person chooses the piece they prefer
- Why is this fair?
- What's the optimal strategy for the cutter?

Let's Try It!

Volunteer Demonstration

- Cutter: Try to make pieces of equal value
- Chooser: Select your preferred piece

Discussion:

- What strategy did the cutter use?
- Was the division fair?

Three People: The Moving Knife Procedure

1. A knife moves slowly across the cake from left to right
2. When any player calls "stop," they receive the piece to the left of the knife
3. The remaining cake is divided among the others

The Dubins-Spanier Protocol (n People)

1. Each person places a mark where they would be satisfied with the piece (from the left edge to their mark)
2. The person who made the leftmost mark gets that piece
3. Repeat with the remaining cake and remaining people

Mathematical Side — Cake Numbers

Definition: The cake number $C(n)$ is the minimum number of cuts needed to divide a cake into n equal pieces.

- $C(1) = 0$
- $C(2) = 1$
- $C(3) =?$
- $C(4) =?$

Calculating Cake Numbers

Formula: $C(n) = \lceil \log_2(n) \rceil$

- $C(2) = \lceil \log_2(2) \rceil = 1$
- $C(3) = \lceil \log_2(3) \rceil = 2$
- $C(4) = \lceil \log_2(4) \rceil = 2$
- $C(5) = \lceil \log_2(5) \rceil = 3$

Why Cake Numbers Matter

- Relate to efficiency in division problems
- Connect to binary numbers and computer science
- Example: To get 5 pieces, we need 3 cuts:
  1. First cut: $1 \rightarrow 2$ pieces
  2. Second cut: $2 \rightarrow 4$ pieces
  3. Third cut: $4 \rightarrow 5$ pieces (by cutting only one piece)

Group Activity (15 minutes)

1. Design a fair division method for 4 people with different preferences
2. Calculate cake numbers for $n = 6, 7, 8,$ and 10
3. Brainstorm real-world applications beyond cake

Example Problem

Four friends want to share a cake:

- Ana loves chocolate (concentrated on left side)
- Ben loves frosting (concentrated on top edge)
- Carlos loves fruit toppings (scattered throughout)
- Dana loves marzipan (concentrated in center)

How would you divide the cake fairly?

Real-World Applications

- Resource allocation problems (land division, inheritance, budgets)
- Computer science (shared computing resources, network bandwidth)
- Conflict resolution (international border disputes, divorce settlements)

Connections to Other Math Topics

- Geometry (areas, volumes)
- Optimization problems
- Game theory
- Combinatorics
- Computer algorithms

Summary

- Fair division is about distributing resources equitably
- Different algorithms work for different numbers of people
- Cake numbers show the minimum cuts needed
- These concepts extend far beyond just cake!

Questions?

Thank you for your participation!

Extension: Banach-Knaster "Last Diminisher" Procedure

1. First person cuts a piece they consider fair (1/n of the cake)
2. Next person can trim if they think it's too large
3. Process continues until the last person
4. Last person to trim (or the first person if no trimming) gets the piece
5. Repeat with remaining $n - 1$ people

Extension: Continuous vs. Discrete Fair Division

- **Continuous**: Division of infinitely divisible resources (like cake)
- **Discrete**: Division of indivisible items (like cookies)
- Different algorithms apply to each case
- Discrete fair division is often more challenging!